

EE3621: Digital Signal Processing

Lecture 5: Z-Transform, ROC & Common Pairs

(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 08:00 (8 mins):** Motivation: Z-transform as a generalization of the DTFT ($z = re^{j\omega}$) and convergence proof.
 - **08:00 – 16:00 (8 mins):** The Region of Convergence (ROC): Does ROC lie inside the unit circle? Causal vs. Anti-causal vs. Two-sided shapes and Stability Comments.
 - **16:00 – 26:00 (10 mins):** Worked Examples: Exp, Sin, Cos, Unit Step, Impulse, and Multi-Pole Multi-Zero System (Magnitude & Phase Computation).
 - **26:00 – 34:00 (8 mins):** Properties of the Z-Transform with full algebraic proofs.
 - **34:00 – 38:00 (4 mins):** Comprehensive Common Z-Transform Pairs Table.
 - **38:00 – 40:00 (2 mins):** Checkpoint and Concept Recap.
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2 Definition & Mathematical Motivation

The Z-transform of a discrete-time sequence $x[n]$ is defined as:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad (1)$$

Expressing z in polar form $z = re^{j\omega}$:

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n} \quad (2)$$

Evaluating $X(z)$ on the unit circle ($|z| = 1 \implies r = 1$) yields the Discrete-Time Fourier Transform (DTFT): $X(z)\Big|_{|z|=1} = X(e^{j\omega})$. The Z-transform series converges absolutely if:

$$\sum_{n=-\infty}^{\infty} |x[n]r^{-n}| < \infty \quad (3)$$

The set of complex values z for which this summation converges defines the **Region of Convergence (ROC)**.

3 Properties of the ROC & Unit Circle Location Clarification

3.1 Does the ROC Lie Inside the Unit Circle?

Whether an ROC lies inside the unit circle depends directly on signal causality and pole magnitude:

1. **Anti-Causal (Left-Sided) Signals:** The ROC is an **interior disk** $|z| < |p|_{min}$. If $|p| < 1$ (e.g., $p = 0.7$), the ROC is $|z| < 0.7$, which **lies entirely INSIDE the unit circle** ($|z| < 1$).
2. **Causal (Right-Sided) Signals:** The ROC is an **exterior domain** $|z| > |p|_{max}$ extending to infinity.
3. **Two-Sided Signals:** The ROC is an **annular ring** $r_1 < |z| < r_2$.
4. **BIBO Stability Criterion:** An LTI system is BIBO stable if and only if the ROC **contains the unit circle** ($|z| = 1$).

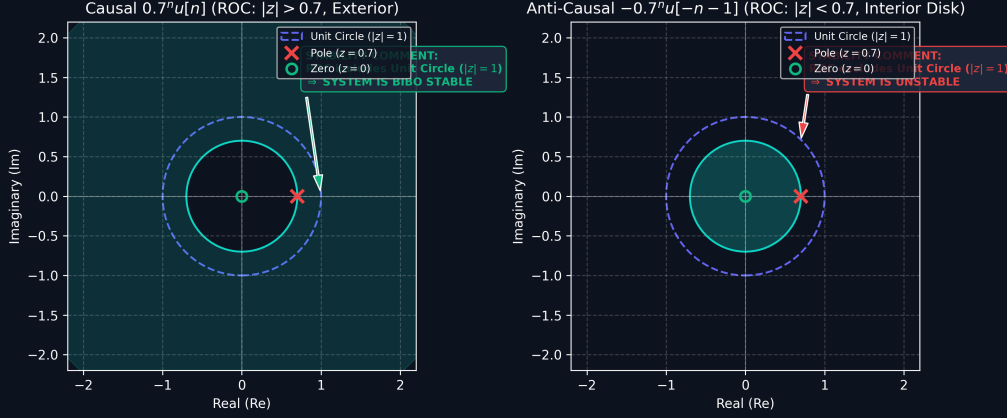


Figure 1: ROC geometries in the z -plane for causal ($|z| > 0.7$) vs. anti-causal ($|z| < 0.7$, inside unit circle) signals with explicit BIBO stability comments.

4 Worked Examples: Signal Variants & Multi-Pole Multi-Zero Systems

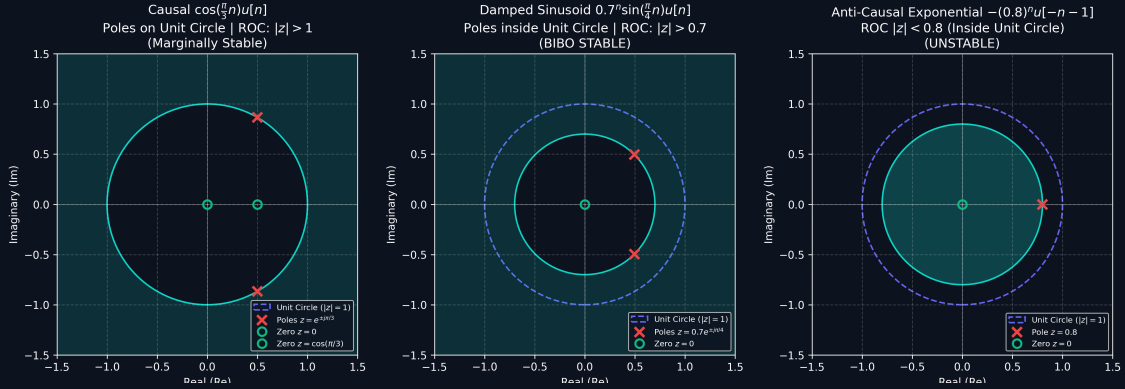


Figure 2: Demonstrative Pole-Zero plots and ROC regions for Causal Cosine, Damped Sinusoid, and Anti-Causal Exponential.

4.1 Example 1: Impulse, Step, Cosine & Damped Sinusoid

- **Unit Impulse** $\delta[n]$: $X(z) = 1$, ROC: All z . **Stable**.
- **Unit Step** $u[n]$: $X(z) = \frac{z}{z-1}$, Pole at $z = 1$, ROC: $|z| > 1$. **Marginally Stable**.
- **Cosine** $\cos(\omega_0 n)u[n]$: $X(z) = \frac{z(z - \cos \omega_0)}{z^2 - 2 \cos \omega_0 z + 1}$, Poles at $z = e^{\pm j\omega_0}$ (on unit circle), ROC: $|z| > 1$. **Marginally Stable**.
- **Damped Sinusoid** $r_0^n \sin(\omega_0 n)u[n]$ ($r_0 = 0.7, \omega_0 = \pi/4$): $X(z) = \frac{0.7 \sin(\pi/4) z}{z^2 - 1.4 \cos(\pi/4) z + 0.49}$, Poles at $z = 0.7e^{\pm j\pi/4}$, ROC: $|z| > 0.7$. **BIBO Stable**.

4.2 Example 2: Multi-Pole Multi-Zero System (ROC, Magnitude & Phase)

Consider $H(z) = \frac{z-0.5}{z^2-0.8z+0.25}$.

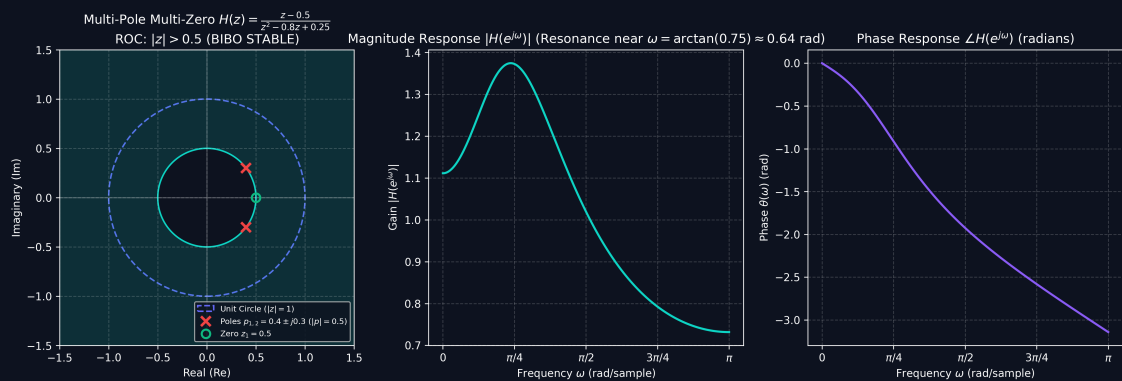


Figure 3: Multi-pole multi-zero system pole-zero plot in z -plane (left), magnitude response $|H(e^{j\omega})|$ (center), and phase response $\angle H(e^{j\omega})$ (right).

1. **Poles & Zeros:** Zero at $z_1 = 0.5$; Complex poles at $p_{1,2} = 0.4 \pm j0.3$ ($|p| = 0.5$).
2. **ROC & Stability:** Causal $\implies |z| > 0.5$. Contains unit circle $|z| = 1 \implies$ **BIBO STABLE**.

3. Magnitude Computation $|H(e^{j\omega})|$:

$$|H(e^{j\omega})| = \frac{|e^{j\omega} - 0.5|}{|e^{j\omega} - (0.4 + j0.3)| \cdot |e^{j\omega} - (0.4 - j0.3)|} \quad (4)$$

At DC ($\omega = 0$): $|H(e^{j0})| = \frac{0.5}{0.45} = 1.111$ (1.09 dB). At resonance ($\omega \approx 0.6435$ rad): $|H(e^{j0.6435})| \approx 1.54$ (3.75 dB). At $\omega = \pi$: $|H(e^{j\pi})| = \frac{1.5}{2.05} = 0.7317$ (-2.71 dB).

4. Phase Computation $\theta(\omega)$:

$$\theta(\omega) = \angle(e^{j\omega} - 0.5) - \angle(e^{j\omega} - (0.4 + j0.3)) - \angle(e^{j\omega} - (0.4 - j0.3)) \quad (5)$$

5 Geometric Vector Method: Magnitude & Phase Computation

Evaluating frequency response by moving along the unit circle $z = e^{j\omega}$:

$$|H(e^{j\omega})| = |C| \cdot \frac{\prod_{k=1}^M |\vec{v}_{zk}(\omega)|}{\prod_{m=1}^N |\vec{v}_{pm}(\omega)|} \quad (6)$$

$$\angle H(e^{j\omega}) = \angle C + \sum_{k=1}^M \angle \vec{v}_{zk}(\omega) - \sum_{m=1}^N \angle \vec{v}_{pm}(\omega) \quad (7)$$

5.1 Case 1: Single Zero System ($H(z) = z - 0.6$)

$|H(e^{j\omega})| = |\vec{v}_{z1}| = \sqrt{1.36 - 1.2 \cos \omega}$, $\angle H(e^{j\omega}) = \text{atan2}(\sin \omega, \cos \omega - 0.6)$. Always **BIBO Stable**.

Frequency ω	Complex z	Zero Vector $\vec{v}_{z1}$	Magnitude $ H(e^{j\omega}) $	Phase $\angle H(e^{j\omega})$
$\omega = 0$	$1 + j0$	$0.4 + j0$	0.4000 (Min)	0.00° (0.000 rad)
$\omega = \pi/2$	$0 + j1$	$-0.6 + j1.0$	1.1662	120.96° (2.111 rad)
$\omega = \pi$	$-1 + j0$	$-1.6 + j0$	1.6000 (Max)	180.00° (π rad)
$\omega = 3\pi/2$	$0 - j1$	$-0.6 - j1.0$	1.1662	-120.96° (-2.111 rad)
$\omega = 2\pi$	$1 + j0$	$0.4 + j0$	0.4000 (Min)	0.00° (0.000 rad)

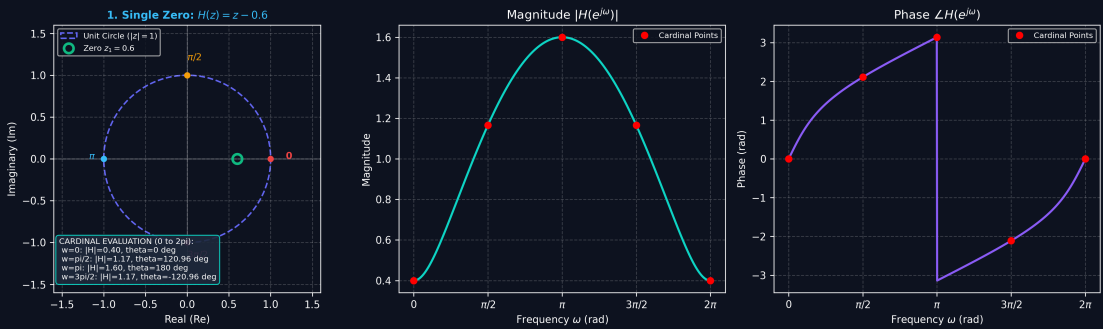


Figure 4: Single zero geometric vector evaluation along the unit circle from $0 \rightarrow 2\pi$.

5.2 Case 2: Single Pole System ($H(z) = \frac{1}{z-0.7}$)

$|H(e^{j\omega})| = \frac{1}{|\vec{v}_{p1}|} = \frac{1}{\sqrt{1.49 - 1.4 \cos \omega}}$. **BIBO Stable** ($|0.7| < 1$).

Frequency ω	Complex z	Pole Vector $\vec{v}_{p1}$	Magnitude $ H(e^{j\omega}) $	Phase $\angle H(e^{j\omega})$
$\omega = 0$	$1 + j0$	$0.3 + j0$	3.3333 (Peak)	0.00° (0.000 rad)
$\omega = \pi/2$	$0 + j1$	$-0.7 + j1.0$	0.8192	-125.00° (-2.182 rad)
$\omega = \pi$	$-1 + j0$	$-1.7 + j0$	0.5882 (Min)	-180.00° ($-\pi$ rad)
$\omega = 3\pi/2$	$0 - j1$	$-0.7 - j1.0$	0.8192	$+125.00^\circ$ ($+2.182$ rad)
$\omega = 2\pi$	$1 + j0$	$0.3 + j0$	3.3333 (Peak)	0.00° (0.000 rad)

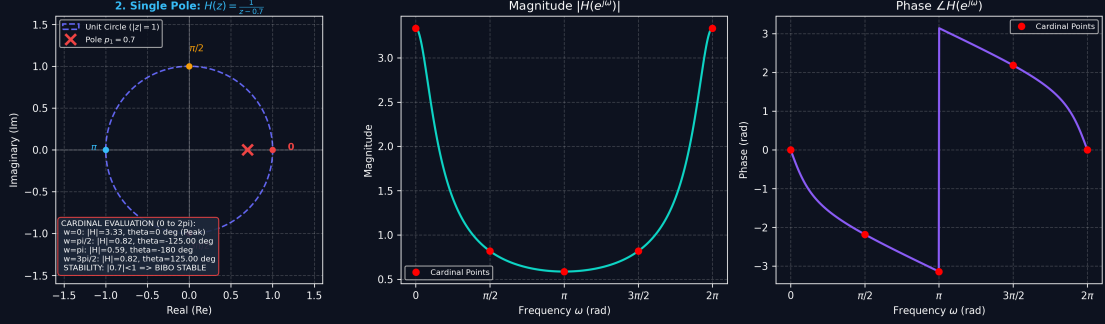


Figure 5: Single pole geometric vector evaluation showing resonance peak at $\omega = 0$.

5.3 Case 3: One Pole + One Zero System ($H(z) = \frac{z-0.8}{z-0.5}$)

$$|H(e^{j\omega})| = \frac{|\vec{v}_{z1}|}{|\vec{v}_{p1}|} = \sqrt{\frac{1.64 - 1.6 \cos \omega}{1.25 - 1.0 \cos \omega}}, \quad \angle H = \theta_{z1} - \phi_{p1}, \quad \text{BIBO Stable } (|0.5| < 1).$$

Frequency ω	Zero Vector $\vec{v}_{z1}$	Pole Vector $\vec{v}_{p1}$	Magnitude $ H(e^{j\omega}) $	Phase $\angle H(e^{j\omega})$
$\omega = 0$	$0.2 + j0$	$0.5 + j0$	0.4000	0.00° (0.000 rad)
$\omega = \pi/2$	$-0.8 + j1.0$	$-0.5 + j1.0$	1.1454	$+12.09^\circ$ ($+0.211$ rad)
$\omega = \pi$	$-1.8 + j0$	$-1.5 + j0$	1.2000	0.00° (0.000 rad)
$\omega = 3\pi/2$	$-0.8 - j1.0$	$-0.5 - j1.0$	1.1454	-12.09° (-0.211 rad)
$\omega = 2\pi$	$0.2 + j0$	$0.5 + j0$	0.4000	0.00° (0.000 rad)

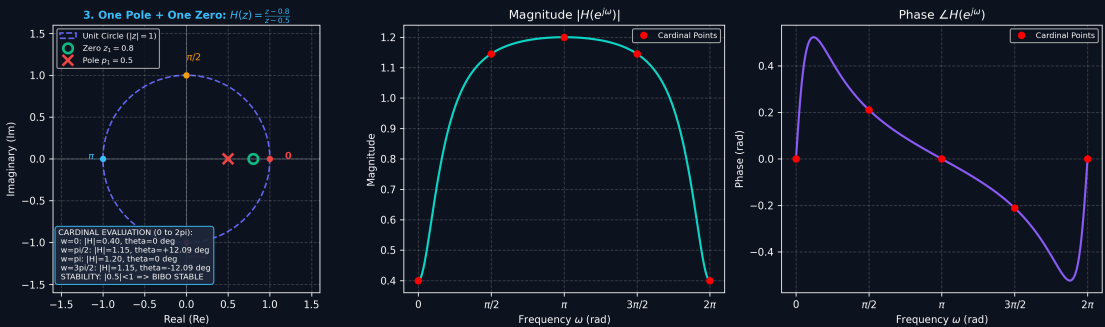


Figure 6: One pole and one zero geometric vector evaluation along unit circle.

5.4 Case 4: Two Poles + Two Zeros System ($H(z) = \frac{z^2-1}{z^2-0.8z+0.25}$)

Zeros at $z = \pm 1$ (notches at $\omega = 0, \pi$), complex poles at $p_{1,2} = 0.4 \pm j0.3$ ($|p| = 0.5$, peak at $\omega \approx 0.6435$ rad). **BIBO Stable**.

Frequency ω	Numerator ($e^{j2\omega} - 1$)	Denominator	Magnitude $ H $	Phase $\angle H(e^{j\omega})$
$\omega = 0$	0	0.45	0.0000 (Notch)	0.00°
$\omega = \pi/2$	-2	$-0.75 - j0.8$	1.8239	$+133.15^\circ$ (+2.324 rad)
$\omega = \pi$	0	2.05	0.0000 (Notch)	0.00°
$\omega = 3\pi/2$	-2	$-0.75 + j0.8$	1.8239	-133.15° (-2.324 rad)
$\omega = 2\pi$	0	0.45	0.0000 (Notch)	0.00°

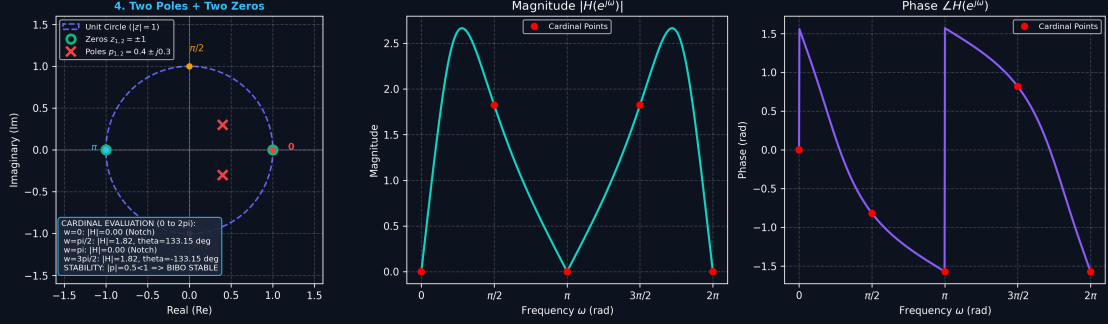


Figure 7: Two poles and two zeros geometric vector evaluation with notches and resonance.

6 Z-Transform Properties & Proofs

For $x[n] \leftrightarrow X(z)$ (ROC R_x) and $y[n] \leftrightarrow Y(z)$ (ROC R_y):

6.1 Time Shifting

$$x[n - n_0] \longleftrightarrow z^{-n_0} X(z) \quad (8)$$

6.2 Scaling in the Z-domain

$$a^n x[n] \longleftrightarrow X(a^{-1}z) \quad (9)$$

6.3 Differentiation in the Z-domain

$$nx[n] \longleftrightarrow -z \frac{dX(z)}{dz} \quad (10)$$

7 Comprehensive Common Z-Transform Pairs Table

Table 1: Comprehensive Z-transform pairs with ROC and Stability comments.

Signal $x[n]$	Z-Transform $X(z)$	ROC	Poles / Zeros	Stability
$\delta[n]$	1	All z	None	Stable
$u[n]$	$\frac{z}{z-1}$	$ z > 1$	Pole: $z = 1$, Zero: $z = 0$	Marginally Stable
$a^n u[n]$	$\frac{z}{z-a}$	$ z > a $	Pole: $z = a$, Zero: $z = 0$	Stable ($ a < 1$)
$-a^n u[-n-1]$	$\frac{z}{z-a}$	$ z < a $	Pole: $z = a$, Zero: $z = 0$	Stable ($ a > 1$)
$nu[n]$	$\frac{z}{(z-1)^2}$	$ z > 1$	Double Pole: $z = 1$	Unstable
$na^n u[n]$	$\frac{az}{(z-a)^2}$	$ z > a $	Double Pole: $z = a$	Stable ($ a < 1$)
$\cos(\omega_0 n)u[n]$	$\frac{z(z - \cos \omega_0)}{z^2 - 2 \cos \omega_0 z + 1}$	$ z > 1$	Poles: $z = e^{\pm j\omega_0}$	Marginally Stable
$\sin(\omega_0 n)u[n]$	$\frac{z \sin \omega_0}{z^2 - 2 \cos \omega_0 z + 1}$	$ z > 1$	Poles: $z = e^{\pm j\omega_0}$	Marginally Stable
$r^n \cos(\omega_0 n)u[n]$	$\frac{z(z - r \cos \omega_0)}{z^2 - 2r \cos \omega_0 z + r^2}$	$ z > r$	Poles: $z = re^{\pm j\omega_0}$	Stable ($r < 1$)
$r^n \sin(\omega_0 n)u[n]$	$\frac{zr \sin \omega_0}{z^2 - 2r \cos \omega_0 z + r^2}$	$ z > r$	Poles: $z = re^{\pm j\omega_0}$	Stable ($r < 1$)

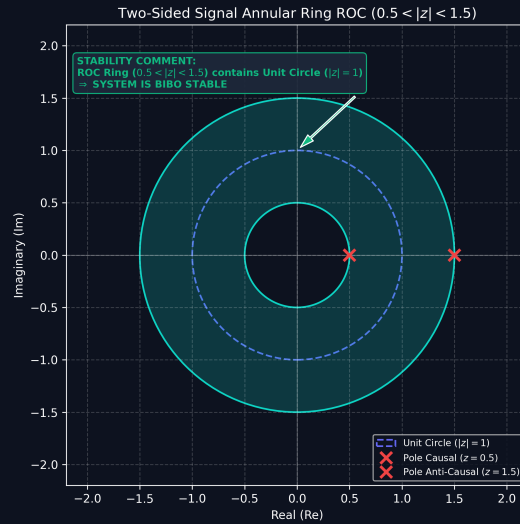


Figure 8: ROC of a stable two-sided sequence, forming an annular ring bounded by pole radii.